Analytic foundations for the time-zero upper-bound construction

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Reading this companion

This companion exposes the analytic dependencies of *Correlated translated probes and the bound* $\Lambda < 0.149396$ *for the de Bruijn–Newman constant* without requiring a reader to locate prose inside nested source archives.

Part A reproduces the pinned all-active probe and time-zero argument from `BCLD_COUPLED_PROBE_MOTION_SOURCE_20260921/PROOF.md`. Part B reproduces the pinned sharp translated-geometry proof supplied with the joint-probe source. Mathematical source text is preserved. Its local equation numbers, historic endpoint values and filenames belong to the indicated part, not to the final publication's scalar update.

The complete numerical parent is regenerated by the standalone replay. The earlier final-time atlas and its 27-volume transport are not dependencies of this construction. Published Polymath and Platt–Trudgian theorems remain external mathematical inputs; the original finite-RH computation is not rerun.

The main paper supplies the correlated two-point improvement and final exact comparison. Neither this text compilation nor an integrity checksum is a proof-assistant verification of the analytic arguments.

Editorial cross-reference for this release

This companion is part of the mathematical submission accompanying version 1.0. The main paper's Introduction gives the previous-bound comparison, including the contextual references to Gomila and Gordon, and distinguishes those references from proof inputs. Its reference list also gives the full journal and DOI details for Polymath and Platt–Trudgian.

Part A, Sections 2–3 contain the finite-head and wall arguments; Sections 4–5 contain the larger-height approximation and derivative error; Sections 6–9 contain the one-point and all-frequency estimates; Sections 10–12 contain the kernel and first-contact comparison. Part B contains the sharp translated geometry. Historical endpoint values in those reproduced parts are intermediate results, not alternative headlines for the present paper. The present endpoint is exactly 18674384228461/125000000000000.

The pre-deposit editorial revision changes this cover guidance and the main paper's exposition and bibliography; it does not change the reproduced mathematical source text, numerical programs or certificates.

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Part I

All-active source estimates and the time-zero barrier

Abstract and verification boundary

A computer-assisted continuation of the author's time-zero argument gives $\Lambda < 0.1511$. The bound is obtained by keeping every active finite-sum term inside the common-phase value/derivative inequality, enclosing time variation after signed coefficient cancellation, and recoupling the translated-probe kernel and self-pair bound to the actual comparison-height enclosure. The exact stronger accepted rational ceiling is regenerated in `ASSEMBLY.json`. The previous 0.151566693470661 endpoint is not a premise.

The proof imports the published Polymath analytic framework and Platt–Trudgian finite-RH theorem. The finite-head wall and every new numerical probe premise are regenerated. No competing unpublished strip, density or jet theorem, no historical final-time atlas, and no open Target-A moment or narrow strip hypothesis is used. It is not proof-assistant formalisation or external peer review. The finite replay checks the explicit numerical premises and algebra; it does not mechanically formalise the analytic arguments below.

1. Statement and complete witness

Use the Polymath normalisation

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du, \quad H_0(z) = \xi((1+iz)/2)/8, \quad \partial_t H_t = -H_{zz},$$

where

$$\Phi(u) = \sum_{n\geq 1}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})e^{-\pi n^2 e^{4u}}.$$

All zeros are counted with multiplicity. Fix

$$X = 6000000185827, \quad N_0 = 690988, \quad v_0 = \log N_0, \quad q_0 = (1000001/1000000)^2, \quad q_e = 10^{-8}.$$

The complete source schedule has 608 closed adjoining intervals $[i/4000, (i+1)/4000]$, $0 \leq i \leq 607$. Probe heights are the exact rationals in `data/PLAN.json`. They start at $109/50 = 2.18$; all are at least 0.69. Six logarithmic-frequency intervals per time cell cover the entire half-line.

Theorem. The analytic reductions in this manuscript, together with the complete regenerated interval and rational checks specified below, prove

$$\boxed{\Lambda \leq U < 1511/10000 = 0.1511.}$$

Here U is the rational output of the upward-enclosed comparison in `ASSEMBLY.json`. A second exact comparison independently meets the stated ceiling using different exponential/logarithm inequalities and endpoint rounding. A failed or partial producer is not a certificate.

The unchanged earlier numerical and analytic sources are retained under `base/`, with their original manifest. The prior adaptive manuscript is also copied to `sources/` to identify every change. Old terminal values and stored success markers are not inputs to the new theorem.

2. Classical framework and the initial finite head

The classical zero strip at time zero is $|\Im z| \leq 1$. De Bruijn contraction in [P, Theorem 3.2] preserves this enclosing strip for $t \geq 0$. The function is real and even, and $H_t(iy) > 0$ for real y, by positivity of its kernel.

[PT, Theorem 1] proves RH through ordinate 3000175332800. The required ordinate is $X/2$, not X. The exact stronger clearance is

$$3000175332800 - (X+1)/2 = 175239886 > 0.$$

The independently regenerated wall in Section 3 proves

$$|H_t(X+iy)/B_t(X+iy)| > 3/4 \quad (0 \le t \le 41/250,\ 0 \le y \le 1). \tag{2.1}$$

It follows that every nonreal zero in this time range has $|\Re z| > X$. Indeed, no zero can enter the finite rectangle through the imaginary axis, its vertical wall or the outer classical strip. A nonreal pair cannot be born inside it from real zeros: the forward Hermite splitting in [P, Proposition 3.1(ii)] gives real, simple nearby branches at a real multiple zero, and real symmetry preserves them. The zero-counting/homotopy argument is local on a compact finite rectangle. This does not assume a positive-time narrow strip.

3. Complete finite-head wall

Write $L = \log(X/(4\pi))$, $\theta_0 = \arg(M_0((1+iX)/2)/M_0((1-iX)/2))$ and $v_n = \log n$. The four time centres are

$$t_j = 41(2j+1)/2000 \quad (0 \le j < 4), \qquad r_t = 41/2000.$$

For every partial prefix $J \le N_0$ and each centre, the producer verifies

$$\sum_{n \le J} n^{-1/2} \cos\left(\frac{Xv_n}{2} - \frac{\theta_0}{2} - \frac{\pi t_j (L-2v_n)}{16}\right) - r_t \sum_{n \le J} \frac{\pi(L-2v_n)}{16\sqrt{n}} > 79/100. \tag{3.1}$$

The second sum encloses the entire time displacement, not a sampled time trace. There are exactly $4N_0 = 2763952$ prefix checks per lane.

The ideal paired model, after dephasing and multiplication by $e^{yL/4}$, has real part equal to twice this cosine sum weighted by

$$w_t(v) = e^{-tv(L-v)/4} \cosh(y(L-2v)/4).$$

Both factors are nonnegative and nonincreasing for $0 \le v \le \log N_0 < L/2$. Finite Abel summation therefore transfers (3.1) to all $0 \le y \le 1$. Undoing the positive factor leaves a floor at least 79/100.

The exact source/gamma displacement and complete unrefined approximation error are bounded by the wall producer. Their sum is less than 1/100, and its exact computed value leaves a floor above 3/4. Both original branches, including the full second exponent, are included. The actual state remains N_0 for all $0 \le t \le 41/250$ on this wall. The checked rational enclosures are in `WALL_256.json` and `WALL_384.json`. The two formulas accumulate the phases differently and both check every partial prefix.

4. Source conventions and a larger-height approximation

For real x, y, t, define

$$a = (1+y-ix)/2, \quad b = 1-a, \quad \alpha(s) = \frac{1}{2s} + \frac{1}{s-1} + \frac{1}{2}\operatorname{Log}\frac{s}{2\pi},$$

$$\beta_t = \alpha + \frac{t}{2}\alpha\alpha', \quad s_U = a + \frac{t}{2}\alpha(a), \quad s_V = b + \frac{t}{2}\alpha(b),$$

$$M_0(s) = \frac{s(s-1)}{16}\pi^{-s/2}\sqrt{2\pi}(s/2)^{s/2-1/2}e^{-s/2}, \quad M_t = M_0 e^{t\alpha^2/4}, \quad B_t = M_t(a), \quad \gamma = M_t(b)/M_t(a).$$

The logarithms are the analytic branches in the respective upper/lower half-planes; there are no poles or branch cuts on the domains used here. Let $b_t(n) = e^{t\log^2 n/4}$ and

$$f_{t,N} = \sum_{n \le N} b_t(n) n^{-s_U} + \gamma \sum_{n \le N} b_t(n) n^{-s_V}, \quad N = \left\lfloor \sqrt{x/(4\pi) + t/16} \right\rfloor. \tag{4.1}$$

This is an **unrefined** model. No corrected terminant is subtracted.

The required probe domains have $0 \le t \le 19/125$, $x \ge X - 2$, and heights listed in the plan between 0.69 and 2.18. Radius-0.01 Cauchy disks remain within $x \ge X - 4$ and $0.54 \le y \le 2.41$. In particular, the real parts of a, b lie in $[-1, 2]$. The theorem [P, 1.3] by itself does not cover the whole height range. The following derivation supplies the required extension.

4.1. A coarse complete Riemann–Siegel remainder

Let $V = x/2$, $V' = V + \pi t/8$, and $A = \sqrt{V'/(2\pi)}$. In [P, Section 6.2] the all-real-σ expansion and Gaussian identity, before the restriction $0 \le \sigma \le 1$ is used to simplify constants, give

$$R_{t,N}(\sigma + iV) = \text{unit phase } e^{t\pi^2/64} M_0(iV')[C_0(p)(1 + O_{\le\epsilon}) + O_{\le\delta_1+\delta_2+\delta_3}]. \tag{4.2}$$

Here $|C_0(p)| \le 1/2$, and, with $D = V - 3$,

$$G_\sigma(c) = \frac{\exp\{5/(6D) + [\sigma^2/D + \sigma c + tc^2/4]/(1 - t/D)\}}{\sqrt{1 - t/D}} = \int_{\mathbb{R}} e^{(u^2+5/6)/D + cu} \frac{e^{-(u-\sigma)^2/t}}{\sqrt{\pi t}} du.$$

For $-1 \le \sigma \le 2$,

$$\epsilon \le e^{(8+t+5/6)/(V-6)} - 1, \quad \delta_1 \le \frac{0.2}{N_0 - 0.865} G_2(\log 9), \quad \delta_2 \le \frac{0.029}{N_0 - 0.353} G_{-1}(-\log 2). \tag{4.3}$$

The first two constants arise from summing every coefficient of the all-σ expansion as in [P, equations (62)–(64)]. They do not discard the Gaussian range in which its real argument lies outside $[-1, 2]$. The displayed Gaussian integral is evaluated exactly. Convexity in σ and direct endpoint comparison give the two indicated maxima; both are increasing in the upper time endpoint.

For completeness, the last sum in [P, (64)] is supported where $u = -r$ and $r \ge V'/\pi - 4 > 100$. Its indices satisfy $k \le r + 4$ and $k \ge V'/\pi$. Use $\Gamma(k/2) \le k! \le e^{k^2}$, $(1.1/A)^k < 1$, and at most $r + 5$ terms. For $0 < t \le 1/5$,

$$t^{-1/2} e^{-(r+\sigma)^2/t} \le \sqrt{5} e^{-5(r-1)^2}.$$

Since $\log(r + 5) \le r$, $\log 2 < 1$, and $D \ge 100$, the complete integrand, including its coefficient and Gaussian normalisation, is at most

$$\exp[-4r^2 + 20r + 11 + (r^2 + 1)/D] \le e^{-3r^2}.$$

Therefore

$$\delta_3 \le \int_{100}^{\infty} e^{-3r^2} dr \le e^{-30000}/600 < 10^{-100}. \tag{4.4}$$

Each time-height row checks $\epsilon/2 + \delta_1 + \delta_2 + 10^{-100} < 1/1000$. Equation (4.2) thus proves the deliberately coarser bound

$$|R_{t,N}(\sigma + iV)| < e^{t\pi^2/64} |M_0(iV')|. \tag{4.5}$$

At $t = 0$ the same assertion follows by the limit of these uniform bounds or directly from the expansion. No assumption about zero locations is used.

4.2. Normalising both remainders

On $-1 \le \sigma \le 2$ and $x \ge X - 4$, direct differentiation gives

$$|\alpha(\sigma \pm ix/2) - (L/2 \pm i\pi/4)| < 10/x, \quad |\alpha'(\sigma \pm ix/2)| < 2/x, \qquad L = \log(x/(4\pi)). \tag{4.6}$$

For example the rational terms contribute at most $3/x$, while the logarithm can be bounded along the segment from $\pm ix/2$ by $2/x$; the constants in (4.6) allow extra room. Also $|\alpha| < L/2 + 1$.

The modulus $|M_0(iV)|$ decreases with V in this range, because $\Im\alpha(iV) = \pi/4 - 1/(2V) - V/(1 + V^2) > 0$. Integrating $\Re\alpha$ from real part zero to $(1 + y)/2$, and using (4.6) for $\Re\alpha^2$, gives a total normalised bound for the two remainders

$$e_C^{\text{broad}} < 4 \exp[-(1 + y)L/4 - tL^2/16 + 1/10]. \tag{4.7}$$

The positive heat correction is at most $t\pi^2/32 < 0.05$; all coordinate terms are less than the remaining slack. The factor four dominates the two complete remainders. There is no partially retained terminant.

5. Physical value and derivative errors on the whole probe domain

For a selected row $[t_L, t_R]$ at height h, use $y_f = h - 0.01$ and $L_* = \log((X-4)/(4\pi))$. Every row checks

$$(1 + y_f)/2 + t_L L_*/8 > 5/4. \tag{5.1}$$

This is a condition on the **union of the selected rectangles**, not on all combinations of time zero and the smallest probe height.

For active indices $\log n \le L/2 + 10^{-12}$, the leading first-branch amplitude is at most $n^{-[(1+y_f)/2+t_L L/8]}$ times a factor below 1.01. The reflected amplitude has ratio at most $n^y e^{-yL/2}$ times the checked source correction. Hence the total complete two-branch mass is below 18. Zeroth, first-log and second-log mass bounds $6, 22, 180$ are individually valid: integrate $v^j e^{-(s-1)v}$ and add its unimodal maximum, using $s > 5/4$.

Proposition 6.1 of [P] is stated for every real σ. From (4.6), its full term error therefore contributes at most

$$e_{AB} < 18(e^d - 1), \quad d = \frac{t_R^2(L_*/2+1)^2/8 + t_R/4 + 1/6}{(X-4)/2 - 3.33}. \tag{5.2}$$

The log-powers divided by x decrease beyond the stated anchor. Equation (4.7) decreases with x, y and t in the required directions. Their maxima are consequently at the endpoints used in the producer.

Freeze the central integer N on a radius-0.01 disk about each probe point. At most one state boundary can be crossed, since the boundary spacing is much larger than 0.02. The complete extra or omitted pair is bounded by

$$e_{\text{pair}} < 3\exp[-(1+y_f)v_0/2 - t_L v_0^2/4 + 10^{-5}]. \tag{5.3}$$

The 10^{-12} activation displacement covers both the heat correction in N and the disk displacement, including a central term momentarily inactive on part of the disk. No sharp-truncation continuity is assumed.

Every row verifies

$$e_{AB} + e_C^{\text{broad}} + e_{\text{pair}} < 10^{-6}. \tag{5.4}$$

The exact finite model with the frozen central N is holomorphic on the disk, as is H_t/B_t. Cauchy's estimate then gives derivative error below 10^{-4} at its centre. This differentiates a bounded analytic discrepancy, not an unknown error term.

Leading source replacement

Set

$$s_U^0 = (1+y)/2 + tL/4 - i(x/2 + \pi t/8), \qquad s_V^0 = (1-y)/2 + tL/4 + i(x/2 + \pi t/8),$$
$$\gamma_0(x,t,y) = \gamma(x,t,0)e^{-yL/2}, \quad |\gamma(x,t,0)| = 1.$$

The leading source is used only with its explicit height derivative at fixed x, t. It is **not** treated as a holomorphic function of z for Cauchy.

Let $\epsilon_s = 100[1 + t_R(L+1)]/x$. Integrating (4.6) along the real source segments and using β_t gives

$$|s_U - s_U^0|, |s_V - s_V^0|, |\log(\gamma/\gamma_0)| < \epsilon_s,$$
$$|s_{U,y} - 1/2|, |s_{V,y} + 1/2|, |(\log\gamma)_y + L/2| < \epsilon_s. \tag{5.5}$$

These follow from $|\alpha| < L/2 + 1$, $|\alpha'| < 2/x$; the integration height is at most 2.41. Each displayed constant is deliberately enlarged. In particular, the complete second exponent, including its κ correction, is accounted for in the source difference.

The complete value and height-derivative changes are bounded by

$$18\epsilon_s(6+22) < 10^{-6}, \qquad 18\epsilon_s(6+22+180)(1+L) < 10^{-4}. \tag{5.6}$$

The producer checks these at $x = X - 4$; the log-powers divided by x decrease thereafter. Products of the small exponential errors are absorbed by the factor 18. Combining (5.4)–(5.6), the complete physical/leading-model errors are $2 \cdot 10^{-6}$ in value and $2 \cdot 10^{-4}$ in height derivative. Finally, (4.6) gives $\Re(\beta_t(a)/2) > L/4 - 10^{-6}$.

6. Complete shared-phase moment inequality

Let $Q = 1 + \sum_j a_j z_j$, $V = \sum_j \lambda_j a_j z_j$, with $a_j, \lambda_j \geq 0$ and $|z_j| = 1$. Add the constant label $a_0 = 1, \lambda_0 = 0, z_0 = 1$. Write $A = \sum_{j>0} a_j$, $B = \sum_j \lambda_j a_j$. For a rational $C > 0$, set

$$D_C = 2 \sum_{i<j} a_i a_j (2C - \lambda_i - \lambda_j)_+.$$

The exact identity

$$C|Q|^2 - \Re(V\bar{Q}) = (1 + A)[C(1 + A) - B] + \sum_{i<j} a_i a_j (2C - \lambda_i - \lambda_j)[\Re(z_i \bar{z}_j) - 1]$$

proves

$$C|Q|^2 - \Re(V\bar{Q}) \geq (1 + A)[C(1 + A) - B] - D_C. \tag{6.1}$$

If $A < 1$ and the right side is positive, then $\Re(V/Q) < C$. No independence, randomness or averaging of the phases is assumed. Signs of coefficients may be absorbed into their unit phases. Coincident frequencies may remain separate labels.

Use the finite Euler factor

$$E_t(s) = \prod_{p=2,3,5} (1 - b_t(p) p^{-s}). \tag{6.2}$$

Its first-branch product head through N_0 has nonnegative coefficients. A direct proof factors an index into labelled prime occurrences and expands

$$b_t(p_1 \cdots p_k) = \prod_i b_t(p_i) \prod_{i<j} [1 + (e^{t \log p_i \log p_j / 2} - 1)].$$

Euler inclusion–exclusion deletes one marked occurrence per selected prime. Only graphs in which every marked occurrence is incident to an edge survive; all their weights are nonnegative and increase with t. Repeated prime factors are retained. This licenses head cancellation only through N_0. The complete product tail is retained with its separate signed labels.

Write $E = \sum_{d|30} m_d d^{-s}$ and let $P_j^{(k)}(\sigma)$ be complete positive prefixes of $b_t(n) n^{-\sigma} (\log n)^k$. If $E_+ = \sum_{d|30} |m_d| d^{-\sigma}$ and $E_1 = \sum_{d|30} |m_d| d^{-\sigma} \log d$, the first-branch amplitude and log-weighted masses are exactly

$$A_U = E_+ P^{(0)}_{N_0} - 1 - 2 \sum_{\mu(d)<0} |m_d| d^{-\sigma} P^{(0)}_{\lfloor N_0/d \rfloor},$$

$$B_U = E_+ P^{(1)}_{N_0} + E_1 P^{(0)}_{N_0} - 2 \sum_{\mu(d)<0} |m_d| d^{-\sigma} [P^{(1)}_{\lfloor N_0/d \rfloor} + \log d \, P^{(0)}_{\lfloor N_0/d \rfloor}]. \tag{6.3}$$

The raw reflected label (d, n) has height frequency

$$\lambda = L - \log n + \log d \geq v_0,$$

so its complete amplitude and weighted masses are added explicitly. The resulting finite core satisfies $Q_y = -V/2$ exactly.

The chosen C is below $17/5$, and $2C < v_0$. Every reflected label and every product-tail label is therefore outside the exceptional set in (6.1). Every possible exception is a pair of first-head indices with product below $e^{2C} < 1024$. Their complete coefficients are regenerated at the upper time endpoint, then weighted at the lower source abscissa. No exceptional pair is omitted.

7. Time variation is bounded after signed cancellation

All bounds in this section are at fixed h, L, N_0. They do not differentiate a changing physical truncation or an unknown approximation remainder. For a label (d, n), $d \mid 30$, $1 \leq n \leq N_0$, put

$$v = \log n, \quad v_d = \log d, \quad q_d = \sum_{p|d} \log^2 p, \quad \kappa_{d,n} = \frac{q_d + v^2 - L(v + v_d)}{4}.$$

Its amplitude has the form $c_{d,n}(h,L)e^{t\kappa_{d,n}}$. Since $L \geq 2v_0$ and $v_d \leq \log 30 < v_0$, one has

$$-K_L \leq \kappa_{d,n} \leq 0, \qquad K_L = \frac{L}{4}(v_0 + \log 30).$$

The first height frequency is $v + v_d$ and the reflected one is $L - v + v_d$. Neither depends on t. The reflected amplitude includes $e^{-hL/2}$; its phase variation in t is immaterial because (6.1) is valid for all phases.

Let $A_0(t,L)$ and $B_0(t,L)$ denote the exact prefix amplitude and weighted mass obtained from (6.3) and the complete reflected labels. They are signed linear combinations of the raw positive label masses after the authorised head cancellation. Let A_{abs} and B_{abs} replace every sign in these linear combinations by an absolute value, including the factor two on every subtracted negative head. The constant in A_0 has zero derivative. Then throughout $[t_L, t_R]$,

$$|A_0''(t,L)| \leq K_L^2 A_{\mathrm{abs}}(t_L, L), \qquad |B_0''(t,L)| \leq K_L^2 B_{\mathrm{abs}}(t_L, L).$$

These follow termwise from $|\kappa| \leq K_L$ and the monotonic decrease of individual absolute label masses in time. No sign of A_0' or B_0' is assumed.

For $t_m = (t_L + t_R)/2$ and $\delta = (t_R - t_L)/2$, Taylor's theorem gives

$$|A_0(t,L) - A_0(t_m,L)| \leq \delta |A_0'(t_m,L)| + \frac{\delta^2}{2} K_L^2 A_{\mathrm{abs}}(t_L, L), \tag{7.1}$$

and the analogous bound for B_0. The actual signed first derivative is formed **before** its absolute value is taken. A midpoint value alone is not a rectangle bound.

The following two exact label identities specify the weighted derivatives. Writing $Q_d = q_d - Lv_d$,

$$(v^2 - Lv + Q_d)(v + v_d) = v^3 + (v_d - L)v^2 + (Q_d - Lv_d)v + Q_d v_d,$$

$$(v^2 - Lv + Q_d)(L + v_d - v) = -v^3 + (2L + v_d)v^2 - [L(L + v_d) + Q_d]v + Q_d(L + v_d).$$

Lane A uses these expanded log moments; lane B differentiates the Euler and prefix factors separately and uses binwise products. Both enclose every integer term. Their signed derivative enclosures must overlap.

The positive prefixes are enclosed on complete integer blocks. Convex exponent minima are included in the reflected blocks. The first-head coefficients are nonnegative, and the absolute tail/reflected labels are positive; consequently A_0 decreases in L. The weighted reflected mass also decreases since $hv_0 > 2$: its derivative in L is bounded using $1/(L - v + v_d) < h/2$. First-head and first-tail frequencies are independent of L. These facts license upper bounds at the left frequency endpoint and lower A_0 bounds at the right endpoint. The infinite frequency interval uses lower mass zero. Equation (7.1), not an unproved monotonicity of a signed time derivative, covers the entire time cell.

The six frequency intervals for $s = L/2 - v_0$ are

$$[0, 1/20],\ [1/20, 3/20],\ [3/20, 2/5],\ [2/5, 1],\ [1, 2],\ [2, \infty).$$

All actual $x \geq X - 2$ lie in their union.

8. Every future activation, including time zero

The old positive-time maximum formula need not be used at $t = 0$. The following integral envelopes work uniformly for the selected schedule, including $t = 0$. Let

$$a_0 = t_L v_0/2,\ b = (1-h)/2,\ b_+ = (1+h)/2,\ c_0 = a_0 + h,\ k = a_0 - b > 0,\ C_0 = e^{-t_L v_0^2/4 + b v_0},\ \delta = 10^{-12}.$$

For $v = \log n = v_0 + w$, an active future term has $0 < w \leq s + \delta$. Dropping $t(w^2/4 - sw/2) \leq t\delta^2/4$, its two amplitudes are bounded by

$$C_0 e^{-a_0 s} e^{bw}, \qquad C_0 e^{-c_0 s} e^{b_+ w}.$$

The corresponding derivative weights are $(v_0 + w)/2$ and $(v_0 + 2s - w)/2$. Time decrease of active amplitudes licenses $t = t_L$.

Define $J_j(b,z)=\int_0^z w^j e^{bw}dw$, $j=0,1$. These have explicit exponential formulas with their continuous values at $b=0$. On a finite interval $l\le s\le u$, a complete value envelope is

$$C_0e^{10^{-5}}[e^{-a_0l}J_0(b,u+\delta)+e^{-c_0l}J_0(b_+,u+\delta)], \tag{8.1}$$

and a derivative envelope is

$$C_0e^{10^{-5}}\left[\tfrac12 e^{-a_0l}(v_0J_0(b,u+\delta)+J_1(b,u+\delta))+e^{-c_0l}((v_0/2+u+\delta)J_0(b_+,u+\delta)-J_1(b_+,u+\delta)/2)\right]. \tag{8.2}$$

For the infinite interval $s\ge l$, replace the first summand in (8.1) by

$$e^{-a_0l}J_0(b,l+\delta)+e^{-kl}/k,$$

and the second by

$$e^{-c_0l}J_0(b_+,l+\delta)+e^{-kl}/k.$$

The corresponding derivative envelope, before $C_0e^{10^{-5}}$, is

$$\tfrac12 e^{-a_0l}(v_0J_0(b,l+\delta)+J_1(b,l+\delta))+e^{-kl}\left(\frac{v_0+l}{2k}+\frac{1}{2k^2}\right)$$

$$+e^{-c_0l}((v_0/2+l+\delta)J_0(b_+,l+\delta)-J_1(b_+,l+\delta)/2)+e^{-kl}\left(\frac{v_0+l+2\delta}{2k}+\frac{1}{2k^2}\right). \tag{8.3}$$

To prove these expressions, interchange the nonnegative integral and the supremum bound, using $s\ge\max(l,w-\delta)$. The reflected weighted exponential decreases with s because $c_0v_0>2$. The remaining integrals are $\int_l^\infty e^{-kw}dw$ and $\int_l^\infty we^{-kw}dw$. The small overlap at $l,l+\delta$ is included twice, safely. Replacing a sum by its adjacent integral costs at most e^{1/N_0} for a reflected coefficient that increases; its logarithmic growth is at most $(h-1)_+/2<1$. The factor $e^{10^{-5}}$ covers this and every stated endpoint correction, as checked in each row. No high-frequency endpoint is left unbounded.

8.1. Activation belongs inside the moment inequality

Let the complete envelopes (8.1)–(8.3) be $A_{\mathrm{act}},D_{\mathrm{act}}$. For each actual source x,t, retain its actual finite integer N and include all labels $N_0<n\le N$ in the leading Euler-multiplied core Q. There is no evaluation of an infinite heat-weighted Dirichlet series. At fixed x,t, the central N stays frozen when taking the height derivative.

Each first-branch activated label has height frequency $\log(dn)>v_0$. Each reflected activated label has frequency

$$L-\log n+\log d\ge v_0-10^{-12}>2C.$$

Thus these labels contribute no exceptional pair in (6.1), even when the other label is the constant. Signs of Euler-divisor terms are absorbed into their unit phases. Possible cancellation with existing tail labels need not be assumed; retaining them separately is a valid, conservative core.

If E_n and E_y bound the Euler value and height derivative, their full contributions satisfy

$$\boxed{A_{\mathrm{new}}\le E_nA_{\mathrm{act}},\qquad B_{\mathrm{new}}\le 2(E_nD_{\mathrm{act}}+E_yA_{\mathrm{act}}).} \tag{8.4}$$

The second term is mandatory: it is the derivative of the Euler factor on the activated sum. Since $Q_y=-V/2$, the weighted mass is twice the absolute height derivative. The proof spends every added label in both the amplitude and weighted moment. It does not declare future activations negligible.

Use the resulting complete masses A,B in (6.1). The lower bound for A may safely remain the prefix lower bound because every additional label amplitude is nonnegative. The only difference outside Q is now the true physical/leading-model error from Section 5:

$$\boxed{R_0=2\cdot10^{-6}E_n,\qquad R_1=2\cdot10^{-4}E_n+2\cdot10^{-6}E_y.} \tag{8.5}$$

These errors remain absolute debits; they are not assigned invented phases or derivative weights to fold them into the core.

9. Complete all-frequency probe lower bounds

For every closed time-frequency rectangle the producer verifies

$$0 \le A_- \le A \le A_+ < 4/5, \quad B \le B_+, \quad (1+A_-)[C(1+A_-) - B_+] - D_C > 0, \quad 0 < C < 17/5.$$

This implies (6.1) for the actual masses: positivity forces $C(1+A_-) > B_+$, and the expression in A is then increasing for $A \ge A_-$. The product head coefficients in D_C are enclosed at the upper time, with weights at the lower real abscissa; graph monotonicity justifies that direction. Every possible exceptional index product is below 1024. Its enumeration bound is confirmed by interval logarithms, not by an unchecked floating comparison.

Put $g = 1 - A_+$. The condition $g > R_0$ proves probe nonvanishing before division. The complete perturbation of the logarithmic derivative is

$$\left| \frac{(Q+R)_y}{Q+R} - \frac{Q_y}{Q} \right| \le \frac{R_1 + B_+ R_0/(2g)}{g - R_0}. \tag{9.1}$$

With the complete Euler logarithmic derivative bound

$$E_\ell = (1/2 + 10^{-5}) \sum_{p=2,3,5} \frac{\log p \, a_p e^{10^{-5}}}{1 - a_p e^{10^{-5}}}, \quad a_p = b_t(p) p^{-\sigma_U^0},$$

one obtains

$$\boxed{S(x,h,t) := -\Im \frac{H_t'}{H_t}(x+ih) > \frac{L}{4} - 10^{-6} - E_\ell - \frac{C}{2} - \frac{R_1 + B_+ R_0/(2g)}{g - R_0}.} \tag{9.2}$$

The physical source displacement, Cauchy derivative error, finite signed Euler tail, all reflected terms and all future truncations are included.

The exact assembly independently reconstructs (7.1), (8.4)–(8.5), the moment cone, and (9.1)–(9.2) from outward rational enclosures. It uses the minimum of all six frequency rows, and then the **weaker** of the two lanes. A small additional downward rounding allowance is spent before extracting the common density floor. Every one of 608 time cells must supply all six rows, including the infinite interval, or assembly fails.

10. Exact kernel catalogue and an attained-height self-pair bound

For $0 \le b \le y$, define

$$K_h(d,b) = \frac{h-b}{d^2 + (h-b)^2} + \frac{h+b}{d^2 + (h+b)^2}.$$

The 13 exact certificates in `data/KERNELS.json` prove

$$\boxed{2K_y(d,b) \ge \lambda \frac{y}{h} \frac{K_h(d+rh,b) + K_h(d-rh,b)}{2}} \tag{10.1}$$

for every real spacing d, every $0 < y \le \rho h$, and $0 \le b \le y$, except the distinguished singular location. Eight certificates are retained; the five new ones all have $\lambda = 2$:

ρ	r	Exact subdivision leaves
23/50	19/40	4
9/20	51/100	3
91/200	49/100	3
7/20	107/200	2
1/5	113/200	1

Clearing positive denominators gives a polynomial after the substitution $D = d^2/h^2$, $v = y^2/h^2$, $w = b^2/y^2$. The transformation $D = z/(1-z)$, multiplied by its positive denominator, compactifies spacing. The exact checker regenerates the polynomial, verifies the complete subdivision tree and all rational Bernstein

coefficients. In total 8,280 coefficients are checked; nonnegative Bernstein basis functions prove the continuous inequalities. This is not a finite grid test. A separate exact counterexample rejects the invalid extension $\rho = 1/2, r = 1/2, \lambda = 2$. The far-spacing asymptotic requires $\lambda \le 2$ for this form of comparison.

For a highest zero at height y, put $q = y^2$ and $v = q/h^2$. Its own pair contributes $h^{-1}s_r(v)$ to the translated average, where

$$s_r(v) = \frac{2(1 + r^2 - v)}{(1 + r^2 - v)^2 + 4r^2 v}.$$

The catalogue provides a rational global cap M_r on $0 \le v \le \rho^2$, verified by the full quadratic numerator, its endpoints and any interior minimum.

The new coupling can improve that cap on the **actual entire range** $0 \le v \le V$, where $V = q_{\text{upper}}/h^2$ is a certified comparison ceiling. Writing $a = r^2$, the sign of $s_r'(v)$ is the sign of

$$(1 + a - v)^2 - 4a(1 + a).$$

If $3a \ge 1$, s_r decreases, and $s_r(0) = 2/(1 + a)$ is valid. If $(1 + a - V)^2 \ge 4a(1 + a)$, it increases throughout $[0, V]$, and $s_r(V)$ is valid. Otherwise the global M_r is used. In every case take the minimum with the verified global cap. An endpoint value is never used when an interior maximum could exceed it.

11. Coupling the complete probe to zero motion

Published zero dynamics give, at a simple highest upper zero,

$$q' = -2 - 4y \sum_{\text{other pairs}} K_y(d_j, b_j).$$

Real zeros are assigned half-pair weight. Imaginary parts of the conjugately paired sums converge absolutely. The own conjugate gives the -2 term and is counted exactly once.

By (10.1), subtracting the **entire** distinguished pair from the translated probe yields

$$q' \le -2 - \frac{2\lambda}{h} q \left[\ell_i - \frac{s_r(q/h^2)}{h} \right].$$

If $M_{r,V}$ is the complete attained-range cap above, define

$$\boxed{c_i = \frac{2\lambda}{h_i} \left(\ell_i - \frac{M_{r,V}}{h_i} \right) > 0. \qquad q' < -2 - c_i q.} \tag{11.1}$$

At each subinterval the exact checker selects only a catalogue entry with $q_{\text{upper}} \le \rho^2 h_i^2$. It maximises the valid rational rate, not a floating source score. Each shift satisfies $rh_i < 2$, so a contact with $x > X$ has both probes in the proved domain $x \ge X - 2$.

The primary comparison divides each source cell into four closed subcells, and recomputes the best valid kernel and attained-range self cap at their boundaries. The source density floor is valid on the whole original cell. No source estimate is interpolated or extrapolated to accomplish this. The alternate comparison uses two subcells. A separate calculation retains one cell and the global self caps to quantify the coupling effect on the same completed source floors.

Global first contact, knots and collisions

The ideal comparison barrier is formed from continuous solutions of $q' = -2 - c_i q$ on adjoining subcells. The stored dyadic values are **upper endpoint enclosures**, not jumps in that barrier. Choosing each next rate using these upper ceilings remains valid for the actual, no-larger barrier. The barrier starts at $q_0 > 1$; an explicit first-cell check keeps it above one for at least 10^{-8} units of time.

Suppose a zero first reaches the ideal barrier. Positive-time confinement [P, Theorem 1.5] bounds every possible nonreal contact in a compact real range; no numerical value of that range is needed. Local persistence and continuity exclude an earlier strict overshoot. Localisation from Sections 2–3 gives $|\Re z| > X$; symmetry

permits an upper-right contact. A multiple zero would split backwards with an upward displacement of order $\sqrt{\epsilon}$ by the Hermite splitting theorem, whereas this positive piecewise-smooth barrier moves by $O(\epsilon)$; that contradicts first contact. A simple contact contradicts (11.1). At a knot the left-hand rate is used. The argument does not differentiate an unattained supremum, assume simplicity of all zeros, or propagate a positive-time result backwards.

12. Exact terminal-time enclosure

For a subcell of width d, the comparison endpoint is

$$q_{\mathrm{next}} = (q + 2/c)e^{-cd} - 2/c.$$

The primary implementation bounds e^{-cd} above by the alternating polynomial through degree 18 ($0 \le cd < 1$), and rounds the endpoint upward to a 96-bit dyadic. The alternate bounds it above by $1/\sum_{j=0}^{12}(cd)^j/j!$ and uses 112-bit upward dyadics.

When a subcell reaches q_e, put $x = c(q - q_e)/(2 + cq_e)$. The exact remaining time is $\log(1 + x)/c$. The primary uses the alternating polynomial through degree 11 as an upper bound for $\log(1 + x)$, $0 < x < 1$. The alternate uses the independent bound $\log(1 + x) \le x - x^2/[2(1 + x)]$. The exact checker verifies the computed upper remaining time fits the selected source subcell. It then adds the ordinary de Bruijn strip-contraction allowance $q_e/2$ and rounds upward to a multiple of 10^{-15}.

Both outputs must be below 0.1511 and the wall horizon 0.164. Their exact fractions, all intermediate ceilings, selected kernels, rates, time joins and terminal calculations are written to `ASSEMBLY.json`. Ordinary floating point is used only for non-authoritative displays and discovery, never to choose acceptance signs or terminal rounding.

13. Verification and limitations

Run `python3.13\-B\RUN.py\-\/-output\/absolute/path/to/new-empty-output` from this scientific source. The runner checks the complete manifests before and after execution, regenerates both wall lanes, all 3,648 probe rectangles per lane, four complete direct finite-sum cross-checks per lane, the separate time-zero xi/digamma check, all 13 kernels and exact structural tests, then reconstructs the numerical and scalar inequalities using integers/fractions. Direct point checks are supplementary; the continuum proof is Sections 4–9.

The transcendental lanes use the same pinned Arb/FLINT library. They use different heat-derivative formulae, integer partitions and accumulation orders, not independent numerical libraries. The exact assembly is separate Python integer/rational arithmetic. Corruption tests reject incomplete rows, invalid exceptions, wrong source identities, missing infinite intervals, negative activation masses and inconsistent derivative caps.

The prior adaptive bound was separately fully rerun during this continuation; its receipt is preserved for provenance. That endpoint and its saved probe values are **not inputs** to the new run. The wall source and published approximation framework are retained, but their necessary finite numerical premises are regenerated.

This result is a further finite-parameter global upper bound. It does not prove $\Lambda < 0.09$ or RH. The schedule and kernel catalogue are not claimed optimal. No worldwide record-ranking claim is made. The analytic reductions are written proofs for external review, not a proof-assistant formalisation.

References

[P] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann xi function, and a new upper bound for the de Bruijn–Newman constant*, arXiv:1904.12438, version 2 (2019). Theorem 3.2; Proposition 3.1; Theorem 1.5; Propositions 6.1–6.2 and the all-real-abscissa Gaussian derivation of Section 6. https://arxiv.org/abs/1904.12438

[PT] D. Platt and T. Trudgian, *The Riemann hypothesis is true up to* $3 \cdot 10^{12}$, arXiv:2004.09765; Bulletin of the London Mathematical Society 53 (2021), 792–797. Theorem 1 supplies the exact endpoint 3000175332800. https://arxiv.org/abs/2004.09765

[Own] The author's adaptive time-zero source, 20 September 2026, source-bound in `SOURCE_BINDINGS.json`; its analytic reductions are restated here. Old numerical endpoint claims are not proof premises.

Part II

Sharp translated-probe geometry

An analytic optimal-translation law for the zero-motion certificate

Dale James Raymond — research continuation, 21 September 2026

0. Scope and status

This note adds a closed-form continuum kernel to the author's own-source `BCLD_LAMBDA_0150989_OWN_PROOF_VERIFIED_20260921` argument. It does not replace, relax, or import new assumptions into the parent's physical approximation, finite-head wall, or uniform source bounds. The new mathematical arguments are in Sections 1–5; the finite application is in Sections 6–8.

The resulting exact scalar ceiling is

$$U = \frac{30168366072429}{200000000000000} = 0.150841830362145 < 0.150842.$$

This conclusion is a corollary of the same written analytic source/global framework as the parent. The full parent numerical production is regenerated in the attached execution, and the new rational comparison rechecks its complete enclosures. Reproduction is not external peer review or a Lean formalisation. No unproved Target-A strip or moment hypothesis, and no competing preprint's functional theorem, is used.

The sharpness claim below concerns precisely a class of one-height, nonnegative symmetric averaging kernels with maximal far-spacing coefficient. It is NOT optimality of the de Bruijn–Newman bound, all translated kernels, signed averages, multiple heights, or the source estimates.

1. Exact kernel and continuum translation

Let

$$K_h(d,b) = \frac{h-b}{d^2+(h-b)^2} + \frac{h+b}{d^2+(h+b)^2}.$$

Set

$$v_* = \frac{1+\sqrt{117}}{58} = 0.2037354107998615\ldots, \qquad \sqrt{v_*} = 0.45137059142113\ldots.$$

Theorem 1. Suppose $h>0$, $0<y<h$, $0\le b\le y$, and $v = y^2/h^2 \le v_*$. Put

$$\boxed{a(y,h) = \sqrt{(h^2-y^2)/3}.}$$

Then for all real d, apart from the distinguished singular location $d=0, b=y$,

$$\boxed{K_y(d,b) \ge \frac{y}{h}\,\frac{K_h(d+a,b)+K_h(d-a,b)}{2}.} \tag{1}$$

The coefficient in the parent's convention $2K_y \ge \lambda(y/h)\cdots$ is exactly $\lambda = 2$, not a coefficient approaching 2 from below.

Complete polynomial proof

Scale $h = 1$. Write

$$D = d^2, \quad V = y^2, \quad B = b^2 = V(1-u), \quad R = a^2 = (1-V)/3, \quad 0 \le u \le 1.$$

The positive denominator at the lower height is

$$D_y = (D + V - B)^2 + 4DB.$$

Let D_4 be the product of the four positive denominators $(d \pm a)^2 + (1 \pm b)^2$. Multiplying $K_y/y - [K_1(d + a, b) + K_1(d - a, b)]/2$ by $D_y D_4$ gives exactly

$$\frac{8(V-4)(V-1)}{81}\, Q(D, u, V),$$

where

$$\begin{aligned} Q = D &\left[9\left(D - \frac{20V-2}{3}\right)^2 + 12(1 + V - 29V^2)\right] \\ &+ uV\left[135D^2 + (72 - 180V)D + 4(1-V)(4-V)\right] \\ &+ u^2V^2\left[135D + 12(1-V)\right] + 9u^3V^3. \end{aligned} \tag{2}$$

All terms are nonnegative: $0 < V \le v_* < 2/5$, $1 + V - 29V^2 \ge 0$, $D \ge 0$, and $u \ge 0$. The prefactor is positive. The denominators are positive except at the explicitly excluded singular location. This proves (1) on its full domain. The code reconstructs the denominator-cleared numerator directly from the four kernels and verifies identity (2) coefficient by coefficient over the rationals. Its supplementary rational-point checks are not the domain proof.

The upper ratio cannot be silently extended

At $B = V$, the polynomial reduces to

$$D\big[9D^2 + (12 - 120V)D + 52V^2 - 68V + 16\big].$$

For $V > v_*$ sufficiently close to v_*, its value at $D = (20V - 2)/3 > 0$ is negative. Thus this particular continuum displacement is not licensed beyond v_*. The checker includes the explicit unsafe choice $V = 21/100, D = 4/5, B = V$, where the cleared numerator is negative. This is a counterexample to the kernel extension, not a physical zeta zero.

2. The self-pair subtraction has a closed form

At a highest zero $x + iy$, its own conjugate supplies the usual vertical velocity term $-1/y$. Its complete conjugate pair also appears in the probe. It must be removed before the other zeros' attraction is credited.

For the displacement in (1), its contribution is

$$\boxed{\frac{K_h(a, y) + K_h(-a, y)}{2} = K_h(a, y) = \frac{6}{h(4 - v)}, \qquad v = y^2/h^2.} \tag{3}$$

This is an equality. Neither the pair nor part of its contribution is omitted.

For a general squared normalised displacement $R \ge 0$, define

$$f_v(R) = \frac{2(1 + R - v)}{(1 + R - v)^2 + 4Rv}.$$

Its denominator is positive for $0 \le v < 1$, and

$$f_v'(R) = \frac{-2\left[R^2 + 2(1-v)R + (1-v)(3v+1)\right]}{\left[(1+R-v)^2 + 4Rv\right]^2} < 0,$$

$$f_v''(R) = \frac{4\left[R^3 + 3(1-v)R^2 + 3(1-v)(3v+1)R + (1-v)(5v^2+10v+1)\right]}{\left[(1+R-v)^2 + 4Rv\right]^3} > 0. \tag{4}$$

Both derivative identities and (3) are reconstructed and checked exactly.

3. Optimality within the precise positive-averaging class

Theorem 2. Fix $0 < y < h$, put $v = y^2/h^2$, and let μ be a compactly supported symmetric probability measure on dimensionless real translations r. Suppose the maximal-coefficient inequality

$$K_y(d,b) \ge \frac{y}{h}\int K_h(d+hr,b)\,d\mu(r) \tag{5}$$

holds for all real d and all $0 \le b \le y$, with the same singular exception. Then its own-pair subtraction satisfies

$$\boxed{\int K_h(hr,y)\,d\mu(r) \ge \frac{6}{h(4-v)}.} \tag{6}$$

For $v \le v_*$, the two-point measure supported at $r = \pm\sqrt{(1-v)/3}$, with equal weights, attains this minimum and obeys (5). Thus it is optimal in this class, including arbitrary finite positive averages and compactly supported continuous positive averages.

Proof. The large-spacing expansions are

$$K_y(d,b) = \frac{2y}{d^2} - \frac{2y(3b^2+y^2)}{d^4} + O(d^{-6}),$$

$$\frac{y}{h}\int K_h(d+hr,b)d\mu(r) = \frac{2y}{d^2} + \frac{2y[-3b^2-h^2+3h^2\int r^2 d\mu]}{d^4} + O(d^{-6}).$$

Compact support justifies termwise integration; symmetry removes odd powers. The inequality implies

$$\int r^2 d\mu \le \frac{1-v}{3}.$$

By the strict convexity and decreasing property in (4), Jensen gives

$$\int f_v(r^2)d\mu \ge f_v\left(\int r^2 d\mu\right) \ge f_v((1-v)/3) = \frac{6}{4-v}.$$

Divide by h. Theorem 1 proves attainment on the stated ratio range. Equality requires the maximal permitted second moment and constant r^2 almost everywhere; under symmetry this is the stated two-point law. □

What the theorem does not exclude

A smaller leading coefficient $\lambda < 2$ changes the trade-off and is not covered by this optimum. Some retained catalogue kernels still win against the new kernel in the actual numerical schedule. Signed probe weights, multiple probe heights, and a joint source bound stronger than the separate uniform floors are also outside the theorem. There is no conclusion of global optimality for the source method or for Λ.

4. Coupling to zero motion

Use the established paired zero dynamics (Polymath [1], Proposition 3.1). At a hypothetical simple highest upper zero $x + iy$,

$$y' = -\frac{1}{y} - 2 \sum_{\text{other conjugate pairs}} K_y(d_j, b_j), \qquad 0 \le b_j \le y.$$

At heights above all these zeros,

$$S(x, h, t) = -\Im \frac{H_t'}{H_t}(x + ih)$$

is their positive pair-kernel sum (real zeros have the usual half-pair weight). Suppose a complete uniform source theorem gives $S(\xi, h, t) > \ell$ at both translated arguments $\xi = x \pm a$. Summing (1), removing the own pair by (3), and writing $q = y^2$, gives

$$\boxed{q' < -2 - \frac{4q}{h}\left[\ell - \frac{6}{h(4 - q/h^2)}\right].} \tag{7}$$

The displacement depends on the contact height, but this causes no derivative term: the kernel and source inequalities are evaluated instantaneously. The argument never differentiates the probes along their moving spatial paths.

For a comparison upper height $\bar{q}$, if

$$0 < \bar{q}/h^2 < 1, \qquad 29(\bar{q}/h^2)^2 - \bar{q}/h^2 - 1 \le 0,$$

then all $0 < q \le \bar{q}$ are licensed, and (3) increases in q. Consequently the exact constant

$$c = \frac{4}{h}\left[\ell - \frac{6}{h(4 - \bar{q}/h^2)}\right] > 0$$

supplies $q' < -2 - cq$ throughout that segment. This is the rate used by the new exact consumer. It competes with every already certified catalogue rate.

5. Global scope and the inherited source domains

The parent uses

$$X = 6000000185827, \quad q_0 = (1000001/1000000)^2, \quad q_e = 10^{-8}.$$

Its finite-head theorem, from its regenerated wall and the published Platt–Trudgian finite-RH input [2], excludes nonreal zeros in $|\Re z| \le X$. The source theorem holds at $x \ge X - 2$, for every time in each specified closed time cell at its prescribed rational height. It includes every future truncation and an infinite final frequency interval. The source proof, including the larger-height approximation, is copied unchanged in the pinned parent archive.

The new displacements satisfy $a < h/\sqrt{3} < 2$, since every parent height is at most 2.18. Thus at any possible right-half-plane contact $x > X$, both translated probes lie strictly inside $x \ge X - 2$. The source error proof's complex-disk padding is unchanged. No new physical source point or height is introduced into the source theorem.

The continuous comparison barrier retains the parent's first-contact proof: initial separation from the classical strip, positive-time confinement, backward Hermite splitting at possible multiple zeros, and the simple-zero velocity contradiction. The comparison stays above q_e, so its square root is Lipschitz. The last positive strip uses the classical contraction cost $q_e/2$. Dyadic stored endpoints bound this continuous barrier; they are not jumps in a zero trajectory.

6. Exact finite application and validation

The new consumer reconstructs the complete source bounds from both freshly regenerated numerical files, including time-centred signed masses, all-active labels, exceptional-pair debit, physical derivative errors, six complete frequency rectangles, and the wall checks. For each matching rectangle it may use the larger of the two valid lower bounds; taking their maximum is a valid consequence of the two proved inequalities. It then takes the minimum over the six frequency rectangles. It does not average numerical enclosures.

The old-kernel-only comparison is rerun as a regression and reproduces exactly

$$0.150988815325516.$$

The new primary calculation uses four substeps per source cell, upward 96-bit dyadic squared heights, and rational upper bounds for exponentials and the terminal logarithm. It returns

$$\boxed{U = 0.150841830362145 < 0.150842.}$$

The independent-formula comparison uses the same four substeps, different exponential/logarithm inequalities and 112-bit endpoint rounding, returning

$$U_{\mathrm{alt}} = 0.150841830364188 < 0.150842.$$

The primary uses 2414 comparison steps over 604 source cells; the continuum kernel wins at 852 steps. It makes no call to a floating ODE solver.

Two conservative diagnostics are retained:

- The alternate formula with only two substeps gives 0.150842140795061, which does NOT prove the headline 0.150842.
- Using the weaker numerical lane on every matching source rectangle gives 0.150855079852812. The finer valid source enclosures are therefore necessary for the sharper headline; it is not separately established by each transcendental lane.

The improvement over the exact parent value is

$$\boxed{0.000146984963371.}$$

This is solely a geometry/self-subtraction improvement at the same schedule and with the same source estimates. There is no new state atlas, new source height, new arithmetic moment estimate, or competing proof premise.

7. The next outside-class opportunity: a joint two-point source

Theorem 2 shows that simply adding more nonnegative shifted probes at one height cannot improve the own-pair subtraction in its precise maximal-leading- coefficient class. The remaining promising gain is not another atomisation of the same source floor: it is to estimate the two physical probes jointly.

For a fixed real displacement a, define

$$\mathcal{H}_{t,a}(z) = H_t(z+a)H_t(z-a).$$

Where its factors are nonzero,

$$\boxed{\frac{S(x+a,h,t)+S(x-a,h,t)}{2} = -\frac{1}{2}\Im\frac{\partial_z \mathcal{H}_{t,a}(x+ih)}{\mathcal{H}_{t,a}(x+ih)}.} \tag{8}$$

Here a is held fixed during differentiation. Differentiating a composite $a = a(h)$ instead would introduce extra terms and is not this identity.

A complete jointly proved lower bound $\ell_{\text{joint}} > \ell$ would insert its gain directly into (7). It may exceed the independent uniform floor because the two worst phases need not occur simultaneously. This is a possibility, not a proven gain for the physical source family.

In a frozen finite exponential packet $Q(x) = \sum_j c_j e^{i\nu_j x}$, the product has the exact symmetrised expansion

$$Q(x+a)Q(x-a) = \sum_{j,k} c_j c_k e^{i(\nu_j+\nu_k)x} \cos((\nu_j - \nu_k)a). \tag{9}$$

This identifies the relevant interference factor. It must be applied to the complete two-branch model and every source/nonlinear freezing error. It does not imply that either the coefficients or the complete logarithmic derivative are positive. Taking absolute values before combining these factors would lose the prospective gain.

A new joint-source producer, or a signed/multiple-height kernel with its additional upper and lower source requirements, is a genuine next research problem. Neither is declared proved here. The 0.09 initial strip and RH remain open.

8. Reproducibility boundary

`check_algebra.py` reconstructs the continuum polynomial, the self-pair identity, its monotonicity/convexity, and an explicit out-of-domain failure using only exact fractions. `assemble_geometry.py` reconstructs all parent numerical bounds and the exact scalar consequence. The full replay command regenerates the parent's wall and both numerical probe lanes before using it.

The transcendental lanes share the pinned Arb/FLINT implementation; they are not independent numerical libraries. The new algebra and scalar checks use standard-library integer/fraction arithmetic. The mathematical extension and optimality results are written proofs, not formally verified Lean theorems.

The package does not claim external peer review, an exhaustive priority audit, a new proof of finite RH, or closure of Target-A. No custom unproved strip or moment assertion is accepted as a successful certificate.

References

[1] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann xi function, and a new upper bound for the de Bruijn–Newman constant*, arXiv:1904.12438v2; especially Proposition 3.1, Theorem 3.2, Theorem 1.5 and Sections 5–6. The parent explains the exact scope of its larger-height approximation rather than extending the numerical statement of Theorem 1.3 by fiat.

[2] D. Platt and T. Trudgian, *The Riemann hypothesis is true up to 3 · 10^12*, arXiv:2004.09765. The finite-RH theorem is a published input; its historical computation is not repeated here.

[3] Dale James Raymond, own-source package `BCLD_LAMBDA_0150989_OWN_PROOF_VERIFIED_20260921.zip`, SHA-256 `18b56253d8bf26c758b4d6b3884de0d5569ea5e5983a10e36797a21794c5d844`. Its unchanged scientific source is embedded as `Parent_Scientific_Source.zip`.